

Internal degrees of freedom in wave-mechanical structure formation: vorticity, spin, and a rotor extension of the Schrödinger–Poisson system

Edward A. Thomson

Working paper, 2026

Abstract

Wave-mechanical methods describe cold and ultralight dark matter as a complex scalar field evolving under the Schrödinger–Poisson equations, with $|\psi|^2$ the mass density and the phase gradient the velocity. This description is intrinsically irrotational and structureless: it admits neither vorticity in the bulk flow nor internal angular momentum for extended objects. This paper sets out a programme to endow wave-mechanical structure formation with internal degrees of freedom. For vorticity we adopt a gravitational vector potential from the weak-field (gravitoelectromagnetic) limit of general relativity, coupled to the wavefunction by minimal coupling. For spin we show that the leading coupling of an extended body to an external gravitational field is through its quadrupole moment and the tidal tensor, yielding a torque that spins up non-spherical objects under purely Newtonian gravity. We argue that the natural internal variable is a rotor (a unit quaternion), of which the two-component Pauli spinor is the axisymmetric special case, and we propose a multicomponent Schrödinger–Poisson system that carries this internal state and reduces to the scalar theory when it is switched off. We give an implementation roadmap and identify the open theoretical problems. The formulation is presented as a direction of travel: the geometric and multipole content is established, while the self-consistent field equations and their coupling constants remain to be derived and tested.

1 Introduction

The wave-mechanical approach to cosmological structure formation replaces the discrete particles of an N -body simulation with a single complex scalar field ψ . Through the Madelung transform [5] the Schrödinger equation is equivalent to a set of fluid equations, so that $|\psi|^2$ plays the role of a mass density and the gradient of the phase plays the role of a velocity field. Widrow and Kaiser [13] proposed using this equivalence as a numerical method for collisionless matter, and the idea has since become central to the study of ultralight or “fuzzy” dark matter, in which the field is physical rather than a numerical device [1, 8, 9]. The same formalism underlies the wave-mechanical perspective on large-scale structure developed by Short [10], Johnston, Lasenby and Hobson [3], and in the present author’s thesis, whose solver and results this paper builds upon.

The strength of the wave-mechanical description is also the origin of the limitation we address here. Because the velocity is the gradient of a scalar phase, $\mathbf{v} = \nu \nabla \arg \psi$, the flow it represents is potential flow: it is irrotational wherever the wavefunction is smooth and non-zero. Vorticity can appear only at the nodes of ψ , where the phase is undefined and the density vanishes. Equally, a single scalar field has no internal orientation: it cannot represent an extended object that carries intrinsic angular momentum, nor the torque that a tidal field exerts on such an object. Yet real structures rotate, haloes acquire angular momentum through tidal torquing,

and galaxies within clusters show alignment effects. If wave mechanics is to be a complete language for structure formation, and not only for the smooth potential-flow regime, it needs a controlled way to carry these internal degrees of freedom.

This paper describes a programme to provide them. It has two strands, corresponding to the two physically distinct notions that a scalar field omits.

- *Vorticity*, meaning circulation in the bulk velocity field. We generate it by enlarging the force law with a gravitational vector potential, taken from the gravitoelectromagnetic (GEM) weak-field limit of general relativity [6], and coupled to the wavefunction by the same minimal coupling used for a charged particle in a magnetic field [12].
- *Spin*, meaning internal angular momentum of an extended object. We show that this arises already under Newtonian gravity, through the coupling of the object’s quadrupole moment to the external tidal tensor, and we carry it in wave mechanics by promoting the wavefunction to a field that also depends on an internal orientation variable.

The central claim on the spin side is geometric. The coupling of an extended body to gravity is to the rank-two tidal tensor, not to a rank-one field, so the natural internal variable is not a two-component spinor but a *rotor*: a unit element of the even Clifford algebra Cl_3^+ , isomorphic to the unit quaternions, which represents a full three-dimensional orientation. The Pauli-like equation that one might write by analogy with quantum spin then reappears as the special case of an axisymmetric body, not as the fundamental description. Recognising this fixes both the degree-of-freedom counting and the choice of coupling.

We are explicit about status. The multipole and tidal-torque content of Sections 4 and 6 is standard and reliable. The gravitoelectromagnetic route of Section 3 rests on an established weak-field theory. The self-consistent multicomponent Schrödinger–Poisson system of Section 7 is a target rather than a result: its internal Hamiltonian, the measure on the orientation space, and the coupling constants remain to be derived, and the reductions to the scalar theory need to be checked against the comoving formulation of the thesis before any implementation. Section 8 sets out how that programme could be carried through.

2 Wave-mechanical structure formation

We recall the scalar system that the extensions of this paper build on. In physical variables a single wavefunction $\psi(\mathbf{x}, t)$ evolves under the Schrödinger–Poisson equations,

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + m \Phi \right] \psi, \quad \nabla^2 \Phi = 4\pi G \rho, \quad \rho = m |\psi|^2. \quad (1)$$

Writing $\psi = \sqrt{\rho/m} \exp(i\theta)$ and separating real and imaginary parts returns the continuity and Euler equations of a fluid with velocity $\mathbf{v} = (\hbar/m) \nabla \theta$ and an additional “quantum pressure” term [5]. The parameter that controls the correspondence is $\nu = \hbar/m$, which sets both the de Broglie scale and, on a grid of N points, the largest velocity the phase can represent before it aliases, $v_{\text{max}} = \nu \pi N$. Resolving the velocity field therefore requires νN to be large enough; this is the origin of the resolution constraint identified in the accompanying code and thesis, and it is the reason the method is exact for smooth potential flow but says nothing, by itself, about rotational or internal motion.

In the rescaled comoving variables used for the cosmological runs (comoving coordinate $\mathbf{y}$, comoving wavefunction χ , comoving potential U , and a dimensionless parameter $\mathcal{L}$), the same system reads, schematically,

$$2i\mathcal{L} [\Omega_{m0} + (1 - \Omega_{m0})a^3]^{1/2} \frac{\partial \chi}{\partial \ln a} = [-\nabla_{\mathbf{y}}^2 + 3\Omega_{m0} \mathcal{L}^2 U] \chi, \quad \nabla_{\mathbf{y}}^2 U = \chi \chi^* - 1, \quad (2)$$

with the Einstein–de Sitter case recovered by $\Omega_{m0} = 1$. Both (1) and (2) are scalar: ψ and χ carry a density and a phase and nothing else. The remainder of this paper asks what minimal additional structure is needed to represent vorticity and spin, and how to add it without disturbing the parts of the theory that already work.

3 Vorticity via gravitoelectromagnetism

Throughout the standard setting the peculiar velocity is generated from a scalar potential and is therefore initially irrotational, $\nabla \times \mathbf{F}_g = \nabla \times \nabla \phi_g \equiv 0$, so Kelvin’s circulation theorem forbids the growth of vorticity in the smooth pre-shell-crossing regime. Within wave mechanics the only candidate sites for circulation are the nodes of ψ , where the density vanishes and the phase is singular. To generate vorticity dynamically, rather than merely to diagnose singular behaviour, the force law itself must be enlarged so that its curl need not vanish.

A phenomenological vector potential would do this but would be arbitrary. A physically motivated route is gravitoelectromagnetism, the weak-field, slow-motion limit of general relativity in which the field equations take a form analogous to Maxwell’s [6]. One introduces a gravito-electric field $\mathbf{E}_g$, reducing to the Newtonian field in the static limit, and a gravitomagnetic field $\mathbf{B}_g$ sourced by mass currents,

$$\nabla \cdot \mathbf{E}_g = -4\pi G \rho_g, \quad \nabla \cdot \mathbf{B}_g = 0, \quad \nabla \times \mathbf{E}_g = -\frac{\partial \mathbf{B}_g}{\partial t}, \quad \nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2} \mathbf{J}_g + \frac{1}{c^2} \frac{\partial \mathbf{E}_g}{\partial t}, \quad (3)$$

with ρ_g the mass density (identified with $m|\psi|^2$) and $\mathbf{J}_g$ the mass-current density (related to the probability current that already defines the velocity). The associated Lorentz-like force is

$$\mathbf{F}_g = m (\mathbf{E}_g + 2 \mathbf{v} \times \mathbf{B}_g), \quad (4)$$

where the factor of two on the gravitomagnetic term reflects the spin-two character of the gravitational field. The gravitomagnetic force acts perpendicular to the velocity; it is not curl-free, so vorticity generation is no longer forbidden.

This field enters a Schrödinger-like evolution by minimal coupling, exactly as for a charged particle in a magnetic field. Watanabe and Tsukada [12] write

$$i\hbar \frac{\partial}{\partial t} \psi = -\frac{\hbar^2}{2m} \left(\nabla - \frac{ie}{\hbar} \mathbf{A} \right)^2 \psi, \quad (5)$$

and show that the evolution can be handled efficiently with splitting operators; a spatially uniform vector potential changes only the phase, while a spatially varying $\mathbf{A}(\mathbf{x})$ also modifies the kinetic operator. In the gravitational case the charge e is replaced by the mass m and $\mathbf{A}$ by the gravitomagnetic potential $\mathbf{A}_g$, with $\mathbf{B}_g = \nabla \times \mathbf{A}_g$.

That the ingredients are already present can be checked without modifying the dynamics. The gravitomagnetic field of a rotating mass distribution is a dipole,

$$\mathbf{B}_g = \frac{G}{2c^2 r^3} [\mathbf{L} - 3(\mathbf{L} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}}], \quad \mathbf{L} = \sum_j m_j (\mathbf{r}_j \times \mathbf{v}_j), \quad (6)$$

and evaluating $\mathbf{L}$ and $\mathbf{B}_g$ on the N -body outputs used elsewhere in the thesis gives a non-zero field whose ratio to the Newtonian force falls with time and is suppressed by v^2/c^2 . The conclusion is not that the effect is dynamically large in a cosmological setting, but that the field is not identically zero in principle, so that a self-consistent GEM extension is well posed. Including it in the equations of motion, whether in an N -body or a wave-mechanical code, is the natural next step for the vorticity strand.

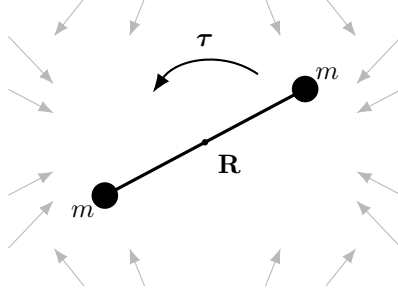


Figure 1: A quadrupolar (dumbbell) super-particle in an external tidal field T_{ij} (grey arrows). The gravitational dipole vanishes, so there is no net dipole force, but the misalignment between the object’s quadrupole tensor and the tidal tensor produces a torque τ about the centre of mass $\mathbf{R}$, spinning the object up under purely Newtonian gravity.

4 Spin of structured objects

The second strand concerns internal angular momentum rather than bulk circulation. A purely scalar Newtonian potential cannot produce a gravitomagnetic force, but it can exert a torque on a body of finite extent when the field varies across it. Spin can therefore arise in a model that remains irrotational at the level of the centre-of-mass flow, which is why it must be treated separately from vorticity.

Consider a super-particle: an unresolved extended object with internal mass distribution $\rho(\mathbf{s})$, where $\mathbf{s}$ is the displacement from the centre of mass $\mathbf{R}$. Following the multipole treatment of an extended body in an external field [2], the interaction energy with an external potential ϕ_{ext} is

$$U(\mathbf{R}) = \int \rho(\mathbf{s}) \phi_{\text{ext}}(\mathbf{R} + \mathbf{s}) d^3s = M \phi_{\text{ext}} + d_i \partial_i \phi_{\text{ext}} + \frac{1}{2} Q_{ij} \partial_i \partial_j \phi_{\text{ext}} + \dots, \quad (7)$$

with

$$M = \int \rho d^3s, \quad d_i = \int \rho s_i d^3s, \quad Q_{ij} = \int \rho s_i s_j d^3s. \quad (8)$$

For a gravitational mass distribution in its own centre-of-mass frame the dipole moment vanishes, $d_i = 0$, because all mass “charges” have the same sign; there is no gravitational analogue of the electric dipole. The first non-trivial coupling of the internal structure to the external field is therefore at quadrupole order, through the tidal tensor $T_{ij} = \partial_i \partial_j \phi_{\text{ext}}$. This is the key physical point: a meaningful spin coupling must be built from the tidal tensor and the quadrupole moment, not from a gravitational dipole that is identically zero.

Writing the local force density as $-\rho(\mathbf{s}) \nabla \phi_{\text{ext}}(\mathbf{R} + \mathbf{s})$ and expanding the gradient to first order in $\mathbf{s}$,

$$\nabla \phi_{\text{ext}}(\mathbf{R} + \mathbf{s}) \approx \nabla \phi_{\text{ext}}(\mathbf{R}) + T \cdot \mathbf{s}, \quad (9)$$

the zeroth-order term produces no torque because the dipole vanishes, and the leading torque about the centre of mass is

$$\tau_i = -\epsilon_{ijk} T_{kl} Q_{jl}. \quad (10)$$

Equation (10) is the clean Newtonian statement of the mechanism: a non-spherical body spins up because its quadrupole tensor is not aligned with the tidal tensor. It carries no spurious numerical factors; the corresponding dumbbell equation of motion in the thesis chapter has been corrected accordingly.

For the simplest illustration, two equal masses at $\mathbf{s}^{(1)} = (+x, 0, 0)$ and $\mathbf{s}^{(2)} = (-x, 0, 0)$ (Figure 1) have zero dipole moment but a non-zero quadrupole $Q_{ij} = 2mx^2 \text{diag}(1, 0, 0)$, and as

the pair rotates the quadrupole stays non-zero. It is the minimal system with non-trivial tidal coupling. For an axisymmetric body with symmetry axis $\mathbf{n}$ one may write

$$Q_{ij} = q n_i n_j, \quad U_{\text{spin}} = \frac{1}{2} q n_i T_{ij} n_j, \quad \boldsymbol{\tau} = -q \mathbf{n} \times (T \cdot \mathbf{n}), \quad (11)$$

which is a single internal degree of freedom, the orientation of $\mathbf{n}$, that can be evolved directly.

5 The Pauli reduction and its limits

Because wave mechanics already accommodates multi-component fields, it is tempting to carry the internal orientation in a spinor and to couple it in the manner of the Pauli equation. Using the isomorphism between the Clifford algebra of three-dimensional space and the complex 2×2 matrices, $\text{Cl}_3 \cong \text{Mat}(2, \mathbb{C})$ with $\mathbf{e}_i \leftrightarrow \sigma_i$ [4], a body axis $\mathbf{d} = d_i \mathbf{e}_i$ maps to the Hermitian matrix $d_i \sigma_i$, and the dipole-order interaction energy takes the form

$$U = \mathbf{B} \cdot \mathbf{S} = \begin{pmatrix} B_z d_z & B_x d_x - i B_y d_y \\ B_x d_x + i B_y d_y & -B_z d_z \end{pmatrix}, \quad B_i = \partial_i \phi, \quad (12)$$

which is Hermitian, as required, and yields a Pauli-like equation

$$i\hbar \frac{\partial}{\partial t} \psi_{\pm} = \left(-\frac{\hbar^2}{2m} \nabla^2 + mV \right) \mathbf{I} \psi_{\pm} - \mu \mathbf{B} \cdot \mathbf{S} \psi_{\pm} \quad (13)$$

for a two-component wavefunction $\psi_{\pm}$, reducing to two independent Schrödinger equations when the coupling is switched off [3]. In this construction the coupling constant is fixed by analogy, $\mu = m$, rather than derived.

There are two reasons not to take this as the fundamental description. First, the gravitational coupling is to the rank-two tidal tensor T_{ij} , not to a rank-one field, so a linear $\mathbf{B} \cdot \boldsymbol{\sigma}$ term is not the natural object; the quadrupole form (10) is. Second, and more concretely, a two-component spinor does not carry enough information to orient a general body. A complex two-spinor has four real parameters; removing the overall norm and phase leaves two physical degrees of freedom, a point on the Bloch sphere, which is a single axis direction $n_i = \xi^\dagger \sigma_i \xi / \xi^\dagger \xi$. That is exactly an axisymmetric body, $Q_{ij} = q n_i n_j$, and no more. A general triaxial object needs three rotational degrees of freedom and, in addition, two trace-free shape parameters to fix the principal quadrupole values. The two-spinor encodes a preferred axis but not a complete body frame, and cannot distinguish rotations about that axis. It is a correct but partial model, adequate for a dumbbell and inadequate for a triaxial halo.

6 Rotor-valued internal states

The variable that does carry a full orientation is a rotor. In three dimensions the even subalgebra Cl_3^+ is isomorphic to the quaternions $\mathbb{H}$, and unit rotors double-cover the rotation group $SO(3)$ [4, 7]. A rotor R satisfies $R\tilde{R} = 1$, where $\tilde{R}$ is the reversion, and it rotates a body-fixed orthonormal frame $\{\mathbf{e}_a\}$ into lab-frame axes

$$\mathbf{E}_a = R \mathbf{e}_a \tilde{R}, \quad a = 1, 2, 3. \quad (14)$$

A body with principal-frame quadrupole $Q^{\text{body}} = \text{diag}(q_1, q_2, q_3)$ then has lab-frame quadrupole

$$Q_{ij}(R) = \sum_{a=1}^3 q_a (\mathbf{E}_a)_i (\mathbf{E}_a)_j = (\mathcal{R}(R) Q^{\text{body}} \mathcal{R}(R)^\top)_{ij}, \quad (15)$$

where $\mathcal{R}(R) \in SO(3)$ is the rotation associated with R , and the interaction keeps the quadrupole form

$$U_{\text{int}}[R] = \frac{1}{2} Q_{ij}(R) T_{ij}. \quad (16)$$

This is the honest generalisation of the axisymmetric model (11): the internal variable is now a full orientation state, and the force and torque continue to descend from the tidal coupling rather than from a guessed linear term. The Pauli two-spinor of Section 5 is recovered by restricting R to rotations of a single axis, that is, by taking $Q^{\text{body}} = \text{diag}(q, 0, 0)$ and reading off $\mathbf{n}$ from the corresponding spinor. The Clifford and Pauli-matrix representations remain useful as computational tools, but they no longer carry the physics unaided: the Hamiltonian comes from (7) and (16), not from the representation.

7 A multicomponent Schrödinger–Poisson system

We now propose how the internal state enters the wave-mechanical evolution. The proposal is schematic: the equations below fix the structure that a self-consistent theory should have, but the internal Hamiltonian, the measure over orientation space, and the coupling constants remain to be derived, and the reductions should be verified against (2).

Promote the scalar field to a field that also depends on an internal orientation coordinate R , and let the density be an integral over that coordinate. In physical variables,

$$i\hbar \frac{\partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + m\Phi + \hat{H}_{\text{int}}(T[\Phi], R) \right] \Psi, \quad (17)$$

$$\nabla^2 \Phi = 4\pi G m \int \Psi^\dagger(\mathbf{x}, t, R) \Psi(\mathbf{x}, t, R) d\mu(R), \quad (18)$$

where the internal Hamiltonian is built from the self-consistent tidal tensor $T_{ij}[\Phi] = \partial_i \partial_j \Phi$ and encodes the quadrupole coupling (16). The essential property is that when the internal state factorises, $\Psi(\mathbf{x}, t, R) = \psi(\mathbf{x}, t) \Xi(R, t)$, with unit norm $\int \Xi^\dagger \Xi d\mu(R) = 1$, and $\hat{H}_{\text{int}}$ is absent or frozen, the density reduces to $\rho = m|\psi|^2$ and (17) collapses to the scalar Schrödinger–Poisson system (1). The construction is an extension of the thesis framework, not a replacement for it.

The same structure written in the comoving variables of (2) adds $\hat{H}_{\text{int}}(\mathcal{T}[U], R)$ inside the bracket, with comoving tidal tensor $\mathcal{T}_{ij}[U] = \partial_{y_i} \partial_{y_j} U$, and replaces $\chi\chi^*$ in the Poisson source by $\int \Psi^\dagger \Psi d\mu(R)$. The scalar comoving system is recovered under the same factorisation and unit-norm condition. Fixing the precise form of $\hat{H}_{\text{int}}$ and $d\mu(R)$, and checking the normalisation against Chapter 5 of the thesis, are the first theoretical tasks of the programme.

8 Implementation roadmap

The formulation is designed to reuse the existing spectral, operator-split solver. The translational part of (17) is unchanged and evolves with the same split-operator kinetic and potential steps. The internal part is a small additional factor in the splitting, acting on the orientation state at fixed position. A practical route is a ladder of increasing self-consistency.

1. *Fixed external tide, one body.* Evolve a single rotor (or its axisymmetric spinor reduction) in a prescribed tidal tensor and verify the torque law (10) against the analytic rigid-body solution. This tests the internal Hamiltonian in isolation.
2. *Self-consistent pair.* Couple two super-particles through the Poisson solve, so that each provides the tidal field of the other, and check angular-momentum bookkeeping and the recovery of the scalar limit when the internal coupling is switched off.
3. *Cosmological field.* Carry the internal state on the full comoving grid, using the multicomponent Poisson source, and measure the statistics of induced spin against the tidal-torque expectation.

At each stage the diagnostics are the conserved quantities (total mass, total angular momentum) and the reduction to the scalar solver when $\hat{H}_{\text{int}} \rightarrow 0$. The vorticity strand of Section 3 can be developed in parallel, since minimal coupling of the gravitomagnetic potential also fits the splitting scheme [12]; combining the two would allow spin-orbit effects to be studied, though that is a later goal.

9 Discussion

The programme is deliberately conservative in its physics and ambitious only in its scope. Each ingredient is either standard (the multipole expansion and tidal torque, the weak-field GEM equations) or a faithful reorganisation of the thesis proposal (the rotor internal state generalising the Pauli spinor). What is new is the claim that these can be assembled into a single multicomponent Schrödinger–Poisson system that carries internal degrees of freedom and reduces cleanly to the scalar theory.

Whether the effects matter dynamically is an open question, and the honest expectation is that they are small in the smooth cosmological regime and largest where tides are strong or objects are compact and rapidly forced. Even where the dynamical effect is small, the framework has diagnostic value: it gives a principled way to read intrinsic spin and its alignment out of a wave-mechanical simulation, quantities that a scalar field cannot represent at all. It also connects the wave-mechanical programme to the broader effort to understand the relationship between the Schrödinger method and its N -body and fluid limits [11, 13], now extended beyond the point-particle idealisation.

10 Conclusion

We have set out a direction for wave-mechanical structure formation: to give it the internal degrees of freedom that a scalar wavefunction lacks. Vorticity is to be generated by a gravitational vector potential from the weak-field limit of general relativity, coupled by minimal coupling. Spin is to be carried by a rotor-valued internal state, coupled to the tidal tensor through the quadrupole moment, with the familiar Pauli spinor recovered as the axisymmetric special case. The two strands come together in a multicomponent Schrödinger–Poisson system that reduces to the scalar theory when the internal state is switched off. The geometric and multipole content of the proposal is established; the self-consistent field equations, their coupling constants, and their numerical behaviour remain to be worked out. The roadmap of Section 8 is how we intend to do that.

References

- [1] W. Hu, R. Barkana, and A. Gruzinov. Fuzzy cold dark matter: The wave properties of ultralight particles. *Physical Review Letters*, 85:1158–1161, 2000.
- [2] J. D. Jackson. *Classical Electrodynamics*. Wiley, 3 edition, 1998.
- [3] R. Johnston, A. N. Lasenby, and M. P. Hobson. Cosmological fluid dynamics in the Schrödinger formalism. *Monthly Notices of the Royal Astronomical Society*, 402:2491–2502, 2010. doi: 10.1111/j.1365-2966.2009.16052.x.
- [4] P. Lounesto. *Clifford Algebras and Spinors*. Cambridge University Press, 2 edition, 2001.
- [5] E. Madelung. Quantentheorie in hydrodynamischer form. *Zeitschrift für Physik*, 40:322–326, 1927.

- [6] B. Mashhoon. Gravitoelectromagnetism: A brief review. *arXiv:gr-qc/0311030*, 2003.
- [7] C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. W. H. Freeman, 1973.
- [8] P. Mocz, M. Vogelsberger, V. H. Robles, J. Zavala, M. Boylan-Kolchin, A. Fialkov, and L. Hernquist. Galaxy formation with BECDM – i. turbulence and relaxation of idealized haloes. *Monthly Notices of the Royal Astronomical Society*, 471:4559–4570, 2017.
- [9] H.-Y. Schive, T. Chiueh, and T. Broadhurst. Cosmic structure as the quantum interference of a coherent dark wave. *Nature Physics*, 10:496–499, 2014.
- [10] C. J. Short. *Large-Scale Structure Formation: A Wave-Mechanical Perspective*. PhD thesis, University of Nottingham, August 2007.
- [11] C. Uhlemann, M. Kopp, and T. Haugg. Schrödinger method as N-body double and UV completion of dust. *Physical Review D*, 90:023517, 2014.
- [12] N. Watanabe and M. Tsukada. Finite element approach for simulating quantum electron dynamics in a magnetic field. *Journal of the Physical Society of Japan*, 69:2962–2965, 2000. doi: 10.1143/JPSJ.69.2962.
- [13] L. M. Widrow and N. Kaiser. Using the Schrödinger equation to simulate collisionless matter. *Astrophysical Journal Letters*, 416:L71–L74, 1993.